

Problem F. Stacks: Balanced Brackets

OS Linux

A bracket is considered to be any one of the following characters: `(`, `)`, `{`, `}`, `[`, or `]`.

Two brackets are considered to be a *matched pair* if the an opening bracket (i.e., `(`, `[`, or `{`) occurs to the left of a closing bracket (i.e., `)`, `]`, or `}`) of the *exact same type*. There are three types of matched pairs of brackets: `[]`, `{}`, and `()`.

A matching pair of brackets is *not balanced* if the set of brackets it encloses are not matched. For example, `{[( )]}` is not balanced because the contents in between `{` and `}` are not balanced. The pair of square brackets encloses a single, unbalanced opening bracket, `(`, and the pair of parentheses encloses a single, unbalanced closing square bracket, `]`.

Some examples of balanced brackets are `[]{}()`, `[({})]{}()` and `{()}{}[]`.

By this logic, we say a sequence of brackets is considered to be *balanced* if the following conditions are met:

- It contains no unmatched brackets.
- The subset of brackets enclosed within the confines of a matched pair of brackets is also a matched pair of brackets.

Given n strings of brackets, determine whether each sequence of brackets is balanced. If a string is balanced, print **YES** on a new line; otherwise, print **NO** on a new line.

Function Description

Complete the `isBalanced` function in the editor below.

`isBalanced` has the following parameter(s):

– *string expression*: a string of brackets

Returns

– *string*: either **YES** or **NO**

Input Format

The first line contains a single integer, n , the number of strings.

Each line i of the n subsequent lines consists of a single string, s , denoting a sequence of brackets.

Constraints

- $1 \leq n \leq 10^3$
- $1 \leq \text{length}(s) \leq 10^3$, where $\text{length}(s)$ is the length of the sequence.
- Each character in the sequence will be a bracket (i.e., `{`, `}`, `(`, `)`, `[`, and `]`).

Input		Output
STDIN	Function	YES
-----	-----	NO
3	t = 3	YES
{[()]}	first expression	
{[()]}	second expression	
{[[[()]]]}	third expression	

Explanation

1. The string `{[()]}` meets both criteria for being a balanced string.
2. The string `{[()]}` is not balanced, because the brackets enclosed by the matched pairs `[()` and `()` are not balanced.
3. The string `{[[[()]]]}` meets both criteria for being a balanced string.